MIL-S-19500/141A(NAVY)
24 February 1966
SUPERSEDING
MIL-S-19500/141(NAVY)
7 February 1961
(See 6.2)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING

TYPE 1N697

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon diode for use in switching circuitry.

1.2 Ratings and characteristics.

V_r	V_{RM} (wkg)	$I_F = I_R = 100mA$ t_{rr}	i_f (surge)	V_F $I_F = 400mA$
v(pk)	v(pk)	nsec	a	---
120	100	100	3.0	1.1V

OPERATING AMBIENT TEMPERATURE: -65° to +150° C

STORAGE TEMPERATURE: -65° to +200° C

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Diodes shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

FSC 5960

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3.5 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Manufacturer's identification.
(b) Country of origin.

3.5.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end.

3.6 Hermetic seal. As part of the manufacturing process, all diodes procured under this specification shall be leak tested as follows. The diodes shall be immersed for a minimum of 1 hour in a pressure tank with a suitable dye such as fluorescein or Rhodamine B penetrant under 150 psig. After washing the diodes, each diode shall be examined under a 20X magnification where Rhodamine B has been used and visually under ultraviolet light where fluorescein dye has been used. Any device exhibiting dye penetration into the seal of the body shall be removed from the lot.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification tests. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.2.1. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1)).

4.2.1 Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Groups A and B.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II and as follows:

4.4.1 Shock. The shock testing apparatus shall be capable of providing shock pulses of the peak acceleration, waveform, and pulse duration specified herein to the body of the device. The acceleration pulse, as determined from the unfiltered output of a transducer with a natural frequency greater than 10,000 cycles per second (cps), shall be a half sine waveform with an allowable distortion not greater than ± 20 percent of the peak acceleration specified herein. The pulse duration shall be measured between the points at 10 percent of the peak acceleration during rise and at 10 percent of the peak acceleration during decay. Absolute tolerance of the pulse duration shall be ± 40 percent of the duration specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

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6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activities:
Navy - SH

User activities:
Navy - WP, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N620(NAVY))

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Table I - Group A inspection.

EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	DETAILS			MIN	MAX	
Subgroup 1			10				
Visual and mechanical examination	2071						
Subgroup 2			5				
Forward voltage	4011	I_F = 400 mAdc		V_F	---	1.1	Vdc
Breakdown voltage	4021	I_R = 5 μAdc		B_V	120	---	Vdc
Reverse current	4016	V_R = 100 Vdc		I_R	---	2.0	μAdc
Reverse current	4016	T_A = 150°C V_R = 100 Vdc		I_R	---	4.0	mAdc
Reverse current at peak reverse voltage	4016	v_r = 120 v(pk)		i_r	---	5.0	μa(pk)
Reverse recovery time	4031	Cond. B I_F = 100 mAdc I_R = 100 mAdc R_L = 1,000 ohms ± 10% C = 100 pf		t_{rr}	---	100	nsec
Capacitance	4001	V_R = 0 Vdc f = 1 mc		C	---	25	pf
Capacitance	4001	V_R = 9 Vdc f = 1 mc		C	---	12	pf

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Table II - Group B inspection.

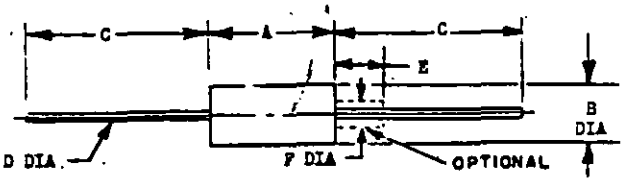
EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHOD	DETAILS			MIN	MAX	
Subgroup 1			10				
Physical dimensions	2066	(See figure 1)					
Subgroup 2			10				
Maximum reverse current	4016	I_R = 0.5 mAdc, 10 ± 1 min.					
Surge current	4066	V_F = 0 Vdc I_f (surge) = 3.0 a $t_p = \frac{1}{120}$ sec 10 surges at 1 surge per minute					
End points:							
Forward voltage	4011	I_F = 400 mAdc		V_F	---	1.1	Vdc
Breakdown voltage	4021	I_R = 5 μAdc		B_V	120	---	Vdc
Reverse current	4016	V_R = 100 Vdc		I_R	---	2.0	μAdc
Subgroup 3			10				
Solderability	2026						
Thermal shock (temperature cycling)	1051	Cond. C					
Thermal shock (glass strain)	1056	Cond. A					
Terminal strength	2036	Cond. A weight = 4 lbs. t = 15 sec.					
Moisture resistance	1021						
End points:							
(Same as subgroup 2)							
Subgroup 4			10				
Shock (See 4.4.1)	2016	Nonoperating 1500 G; t = 0.5 msec; 5 blows each orientation X_1, Y_1, and Y_2					
Vibration, variable frequency	2056						
Constant acceleration	2006	Nonoperating 20,000 G; X_1, Y_1, and Y_2 orientations					
End points:							
(Same as subgroup 2)							

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Table II - Group B inspection (cont'd).

EXAMINATION OR TEST	MIL-STD-750		LTPD	SYMBOL	LIMITS		UNIT
	METHODS	DETAILS			MIN	MAX	
Subgroup 5			10				
Terminal strength	2036	Cond. E					
Subgroup 6			10				
Salt atmosphere	1041						
Subgroup 7			10				
Thermal resistance	4081			θ_{J-A}	---	0.5	°C/mW
End points: (Same as subgroup 2)							
Subgroup 8			10				
High-temperature life (nonoperating)	1031	T_A = 200°C					
End points:							
Forward voltage	4011	I_F = 400 mAdc		V_F	---	1.2	Vdc
Breakdown voltage	4021	I_R = 10 uAdc		B_V	120	---	Vdc
Reverse current	4016	V_R = 100 Vdc		I_R	---	4.0	mAdc
Subgroup 9			10				
Steady state operation life	1026	$T_A = 25^{+5}_{-0}$ °C f = 60 cps v_r = 120 v(pk) i_f = 500 ma(pk)					
End points:							
(Same as subgroup 8)							

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DIMENSIONS			NOTES
LTR	INCHES		
	MIN	MAX	
A	.230	.300	
B	.085	.107	
C	1.500	1.750	
D	.018	.022	2
E	---	.100	
F	---	.075	

NOTES:

1. See 3.5 for marking requirements.
2. Both leads shall be within the specified limit.

Figure 1 - Dimensions for semiconductor device, diode, type 1N697.

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